

Review article

Molecular level studies on interfacial hydration of zwitterionic and other antifouling polymers in situ[☆]Chuan Leng^a, Shuwen Sun^a, Kexin Zhang^a, Shaoyi Jiang^{b,*}, Zhan Chen^{a,*}^a Department of Chemistry, University of Michigan, Ann Arbor, MI 48109, United States^b Department of Chemical Engineering, University of Washington, Seattle, WA 98195, United States

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ABSTRACT

Antifouling polymers have wide applications in biomedical engineering and marine industry. Recently, zwitterionic materials have been reported as promising candidates for antifouling applications, while strong hydration is believed to be the key antifouling mechanism. Zwitterionic materials can be designed with various molecular structures, which affect their hydration and antifouling performance. Although strong hydration has been proposed to occur at the material surfaces, probing the solid material/water interfaces is challenging with traditional analytical techniques. Here in this review, we will review our studies on surface hydration of zwitterionic materials and other antifouling materials by using sum frequency generation (SFG) vibrational spectroscopy, which provides molecular understanding of the water structures at various material surfaces. The materials studied include zwitterionic polymer brushes with different molecular structures, amphiphilic polymers with zwitterionic groups, uncharged hydrophilic polymer brushes, amphiphilic polypeptoids, and widely used antifouling material poly(ethylene glycol). We will compare the differences among zwitterionic materials with various molecular structures as well as the differences between antifouling materials and fouling surfaces of control samples. We will also discuss the effects of pH and biological molecules like proteins on the surface hydration of the zwitterionic materials. Using SFG spectroscopy, we have measured the hydration layers of antifouling materials and found that strong hydrogen bonds are key to the formation of strong hydration layers preventing protein fouling at the polymer interfaces.

Statement of Significance

Antifouling polymers have wide applications in biomedical engineering and marine industry. Recently, zwitterionic materials have been reported as promising candidates for antifouling applications, while strong hydration is believed to be the key antifouling mechanism. However, zwitterionic materials can be designed with various molecular structures, which affect their hydration and antifouling performance. Moreover, although strong hydration has been proposed to occur at the material surfaces, probing the solid material/water interfaces is challenging with traditional analytical techniques. Here in this manuscript, we will review our studies on surface hydration of zwitterionic materials and other antifouling materials by using sum frequency generation (SFG) vibrational spectroscopy, which provides molecular understanding of the water structures at various material surfaces. The materials studied include zwitterionic polymer brushes with different molecular structures, amphiphilic polymers with zwitterionic groups, uncharged hydrophilic polymer brushes, amphiphilic polypeptoids, and widely used antifouling material poly(ethylene glycol). We will compare the differences among zwitterionic materials with various molecular structures as well as the differences between antifouling materials and fouling surfaces of control samples. We will also discuss the effects of pH and biological molecules like proteins on the surface hydration of the zwitterionic materials. All the SFG results indicate that strongly hydrogen-bonded water at the materials' surfaces (strong surface hydration) is closely correlated to the good antifouling properties of the materials. This review will be widely interested by readers of *Acta*

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Biomaterialia and will impact many different research fields in chemistry, materials, engineering, and beyond.

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1. Introduction

Polymers are widely used as biomedical materials and marine antifouling coatings [1–4]. For both applications, antifouling property of a polymer is desired. It is generally agreed that if the adsorption amount of molecules (e.g., proteins) on a surface is less than 5 ng/cm², the surface is considered as nonfouling [5]. Extensive research has been performed to design, synthesize, characterize, and apply nonfouling (or antifouling – please notice that nonfouling and antifouling have a similar meaning in this article) polymeric materials [6–8]. Many polymers that exhibit excellent nonfouling properties have been developed and widely used, such as poly(ethylene glycol) (PEG) [9–12], hydrogels [13–15], amphiphilic materials [16–22], and zwitterionic polymers [5,23–26], etc.

For nonfouling materials, whereas different molecular mechanisms have been proposed to interpret their nonfouling performance, it is widely believed that for hydrophilic polymers and some amphiphilic polymers, strong surface hydration is a key contributor to their nonfouling property [27]. A highly hydrated polymer surface can bind water molecules so strongly that other molecules and organisms cannot replace these interfacial water molecules for fouling to occur. Although this interpretation is widely accepted, direct evidence to show the strong interaction between nonfouling polymers and water molecules has rarely been reported. This is because it is challenging to probe the polymer/water interface in situ using traditional surface sensitive analytical techniques. Most of such techniques either require high vacuum to operate (e.g., X-ray spectroscopy, electron microscopy), or do not provide molecular information (e.g., atomic force microscopy).

Recently, sum frequency generation (SFG) vibrational spectroscopy has been developed into a powerful tool to investigate molecular structures of surfaces and buried interfaces, including solid/liquid interfaces in situ in real time [28–30]. In particular, extensive research has been performed to study molecular behavior of water molecules on surfaces and at interfaces [31–33]. Advanced SFG methods such as phase sensitive detection and pump-probe ultrafast SFG as well as simulations of SFG spectra have been used to study detailed structure and fast dynamics of interfacial water molecules [34–37].

In the past years, we have extensively used SFG spectroscopy to study polymer/water interfaces and biological interfaces (e.g., pep-

tides and proteins at solid/liquid interfaces) [38–44]. We revealed different surface structures of polymers in air and water [45–49], and investigated water structures at the polymer/water interfaces [50–52]. The polymers we investigated include antifouling materials such as zwitterionic polymers [50–53], PEGs [50,51,54,55], amphiphilic polymers [53,56,57], and fouling release polymers such as poly(dimethyl siloxane) [58–62]. Herein, we will review our SFG studies on interfacial water structures at various antifouling polymer surfaces. The polymers we examined include zwitterionic polymers (with different negatively charged functional groups, varied distances between positive and negative charges (carbon spacer length), and different side chains), poly(N-hydroxyalkyl acrylamide) (PolyHAAAs) (with different lengths of side chains), and polypeptoids (with systematically varied side-chain sequences). These studies provide direct evidence that strong surface hydration is critical to the antifouling property of a polymer material. In addition to the antifouling polymers, some control polymers which show fouling activity were also investigated, which revealed their weaker surface hydration. Therefore, these studies provide a general and direct correlation between strong surface hydration and good antifouling property.

2. SFG vibrational spectroscopy

SFG vibrational spectroscopy is an intrinsically surface-sensitive and in situ vibrational spectroscopic method [63,64]. An SFG process involves two input photons and one outgoing photon. The outgoing photon has the sum frequency of the input photons (Fig. 1A). Therefore, SFG signal is related to two input optical beams by a second-order nonlinear optical process. Experimentally, visible and infrared (IR) input lasers penetrate a CaF₂ or SiO₂ substrate (window or prism), and overlap spatially and temporally at the polymer surface in air, water, or an aqueous solution (Fig. 1B). The wavelength of the visible beam is 532 nm, where the IR beam is frequency tunable. The reflected SFG signal is collected by a monochromator along with a photomultiplier tube as a function of the wavenumber of the IR beam. All SFG spectra presented in this work were collected using the ssp (s-polarized sum frequency output, s-polarized visible input, and p-polarized IR input) polarization combination.

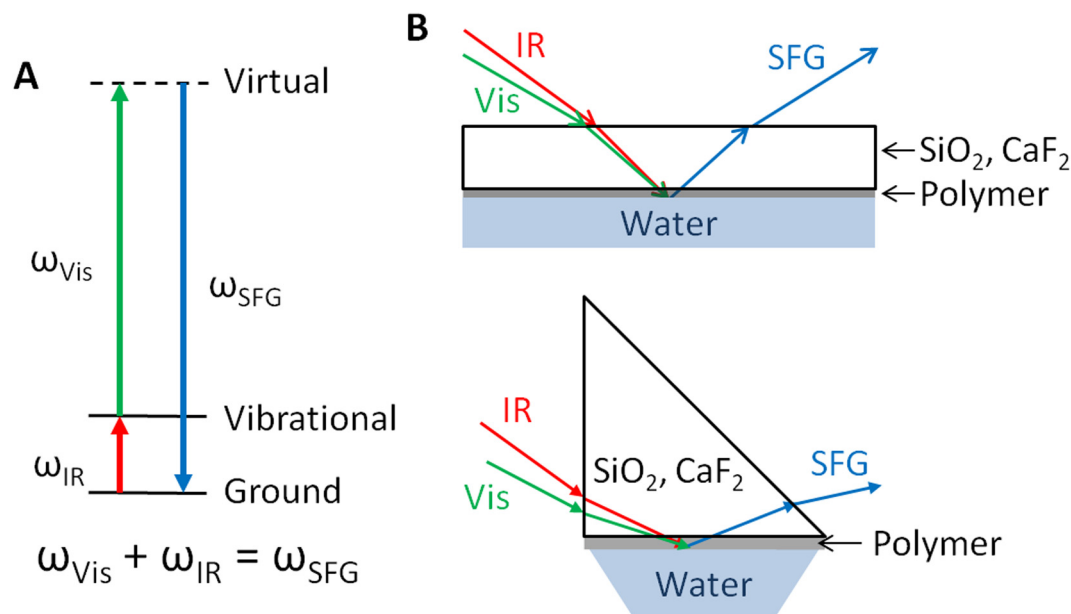


Fig. 1. (A) SFG energy level diagram; (B) experimental scheme of SFG vibrational spectroscopy. Top: window geometry; Bottom: prism geometry.

SFG is surface sensitive because of its selection rule. According to the selection rule, SFG signal intensity is proportional to the square of the second order nonlinear optical susceptibility, $\chi^{(2)}$, of the material under the electric dipole approximation. As a polar third-rank tensor, $\chi^{(2)}$ changes sign under the inversion operation: $\chi^{(2)}(r) = -\chi^{(2)}(-r)$. For materials with inversion symmetry, $\chi^{(2)}(r) = \chi^{(2)}(-r)$. The only possible solution for the above two equations is $\chi^{(2)} = 0$. Therefore, for materials with inversion symmetry, no SFG signal can be generated. Most solid polymer systems do have inversion symmetry, because the functional groups in the bulk of these polymeric materials are generally randomly oriented; therefore they would not generate SFG signals. However, because inversion symmetry is broken at the surface/interface, for molecules or functional groups on polymer surfaces or at polymer interfaces, $\chi^{(2)}(r)$ does not equal to $\chi^{(2)}(-r)$. Here, $\chi^{(2)}$ can be non-zero, and SFG signals can be detected from surfaces and interfaces. Due to this selection rule, SFG is intrinsically surface/interface sensitive. The surface/interface sensitivity of SFG is not determined by the penetration depth of the input laser beams into the sample. Even when the input laser beams can penetrate the entire sample, only molecules or functional groups on the surfaces or at the interfaces contribute to the signals due to the SFG selection rule.

As signals detected in other vibrational spectroscopic methods, SFG vibrational peaks can be assigned to different vibrational modes of various functional groups in the molecules. Since SFG is surface/interface sensitive, SFG spectra can be used to determine what types of molecules/functional groups are present on the surface/at the interface according to the peak centers of the vibrational bands in the spectra. The SFG signals detected from water molecules at interfaces usually contain two O–H stretching peaks centered around 3200 cm^{-1} and 3400 cm^{-1} , generated by strongly hydrogen bonded water and weakly hydrogen bonded water, respectively [65]. SFG intensity is correlated to the ordering of the dipole moments of the interfacial functional groups. Strong SFG signals of water indicate strong ordering of water molecules at the interface. From the water spectra collected from an antifouling material/water interface, we can deduce the hydration strength of the material surface. Strong 3200 cm^{-1} signal indicates a strong surface hydration.

3. Zwitterionic polymers

Recently zwitterionic polymer brushes prepared on gold or glass substrates have been reported as promising materials to resist biofouling in complex environments [66,67]. It is believed that the non-fouling ability of zwitterionic polymers is closely related to an electrostatically induced hydration layer at the surface: The tightly bound water molecules form a physical and energy barrier to prevent non-specific adsorption of biomolecules and organisms [27]. However, zwitterionic materials can be designed with different molecular structures which may change their non-fouling properties. For example, poly(carboxybetaine acrylamide)s (CBAA)s have been synthesized with one to five carbon atoms (CBAA1–5) between the positively charged quaternary amine and the negatively charged carboxyl groups. When the

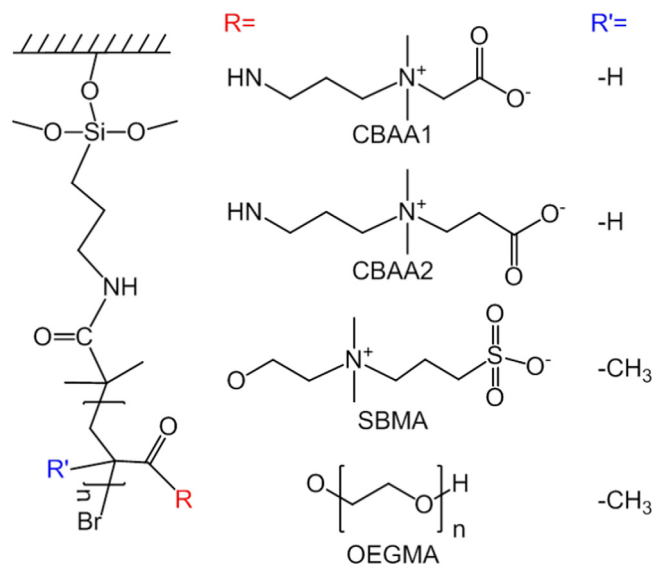


Fig. 2. Molecular structures of nonfouling zwitterionic and PEG polymer brushes (pCBAA1, pCBAA2, pSBMA, and pOEGMA) synthesized on silica substrates for SFG studies [50–52]. (Reproduced with permission from J. Phys. Chem. C 119 (2015) 8775–8780. Copyright 2015 American Chemical Society.)

number of such carbon atoms is greater than two, protein adsorption will start to occur on CBAA, particularly in complex media [68]. In addition, environmental parameters like solution pH and ionic strength can alter the non-fouling performance. Carboxybetaine polymers can resist non-specific protein adsorption at protein solution pH > 5, but protein adsorption occurs at low protein

solution pH values. Meanwhile anti-fouling performance of sulfobetaine materials is insensitive to protein solution pH [69]. As surface hydration is proposed to play a key role to prevent biofouling, in situ examination of the polymer/water interface is highly desired.

3.1. Poly(carboxybetaine acrylamides) (pCBAs) and Poly(sulfobetaine methacrylate) (pSBMA)

As shown in Fig. 2, pCBAA1, pCBAA2, and pSBMA can be synthesized on a silica substrate through atom transfer radical polymerization (ATRP) [24,68]. To understand the differences among zwitterionic materials with different molecular structures, we compared pCBAA1 and pCBAA2 – they have different carbon spacer lengths between the charged groups. We also compared pCBAs and pSBMA – they have different negatively charged groups, carboxylate vs. sulfonate. Although pSBMA has one more carbon atom between the charged groups than pCBAA2, our research showed that the negatively charged groups played a more important role in determining their differences. Furthermore, we aimed to understand the differences between zwitterionic polymers and widely used PEG materials. The latter can also be synthesized as a polymer brush, namely poly(oligo ethylene glycol methacrylate) (pOEGMA), the molecular structure of which is also shown in Fig. 2 [10].

3.1.1. Strong hydration

Fig. 3 shows that each of the SFG water spectra taken at the three zwitterionic polymer/water interfaces exhibits a strong peak

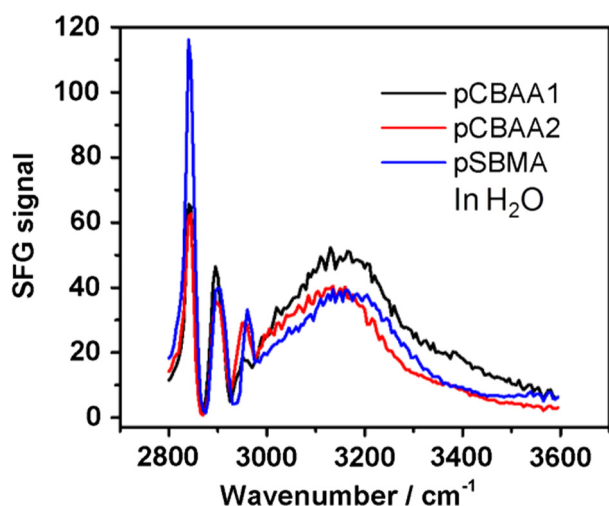


Fig. 3. SFG spectra of pCBAA1, pCBAA2, and pSBMA in water [52]. (Adapted with permission from *J. Phys. Chem. C* 118 (2014) 15840–15845. Copyright 2014 American Chemical Society.)

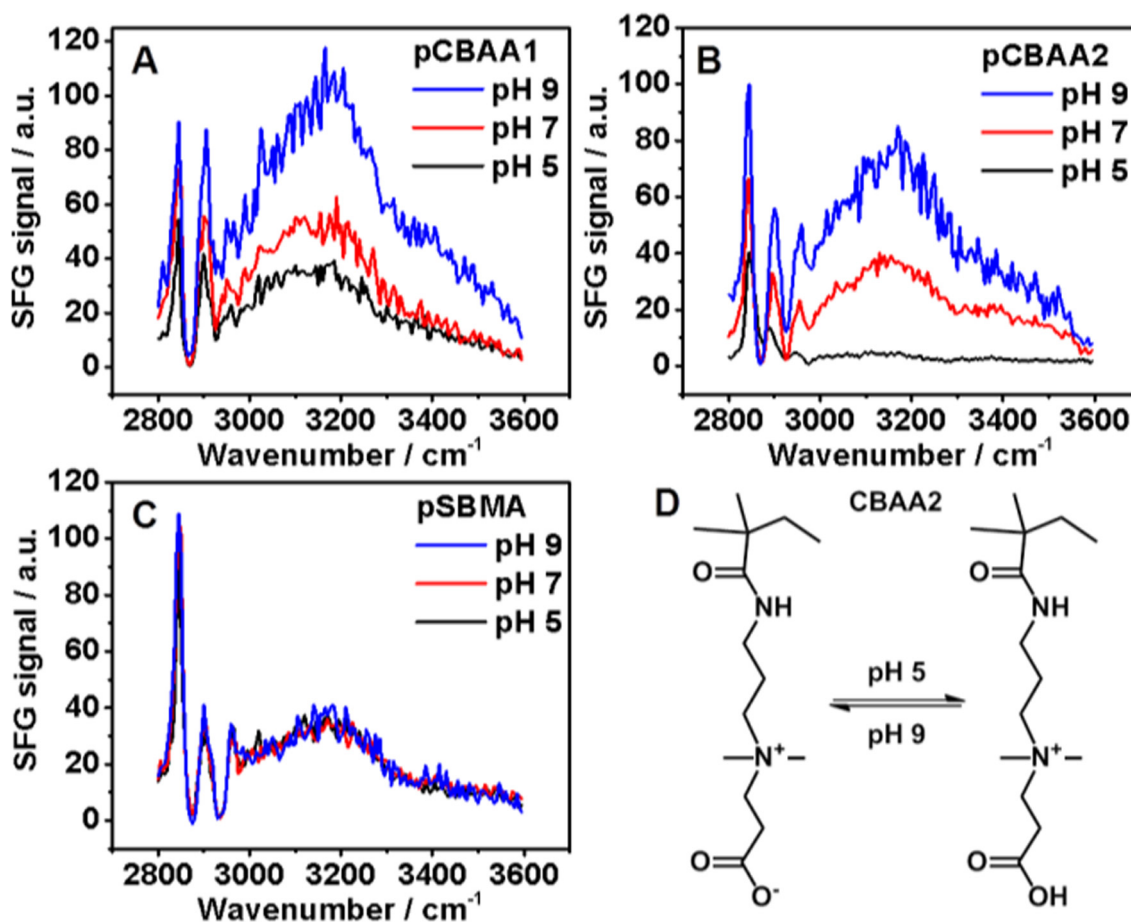


Fig. 4. SFG spectra of (A) pCBAA1, (B) pCBAA2, and (C) pSBMA in contact with aqueous phase at pH of 5, 7, and 9. (D) Protonation and deprotonation of a CBAA2 unit [52]. (Reproduced with permission from *J. Phys. Chem. C* 118 (2014) 15840–15845. Copyright 2014 American Chemical Society.)

at 3200 cm^{-1} [52], which is assigned to strongly hydrogen-bonded water molecules [65], indicating that all the three zwitterionic material surfaces are strongly hydrated. Therefore, we believe that here the strong surface hydration can be correlated to the nonfouling properties of the zwitterionic polymers.

3.1.2. Different carbon spacer lengths: pCBAA1 and pCBAA2

The property difference between pCBAA1 and pCBAA2 is reflected in their responses to the change of pH of the aqueous phase in contact with the polymer (Fig. 4) [52]. As the pH of the

aqueous phase decreased from 9 to 5, for both pCBAA1 and pCBAA2, the SFG water signal decreased, indicating reduced ordering of the interfacial water molecules. At pH 5, water signal was still observable for pCBAA1 (Fig. 4A). However, no water signal was observed at the pCBAA2 surface at this pH, indicating the complete loss of water ordering at the pCBAA2 interface (Fig. 4B). The different hydration behaviors between pCBAA1 and pCBAA2 at a low pH are owing to their different pKa values. The CB-2 molecule (the zwitterionic portion of pCBAA2) has a higher pKa than CB-1 (the zwitterionic portion of pCBAA1) [70], so CB-2 is easier to be protonated (Fig. 4D). Protonation of the carboxylate group weak-

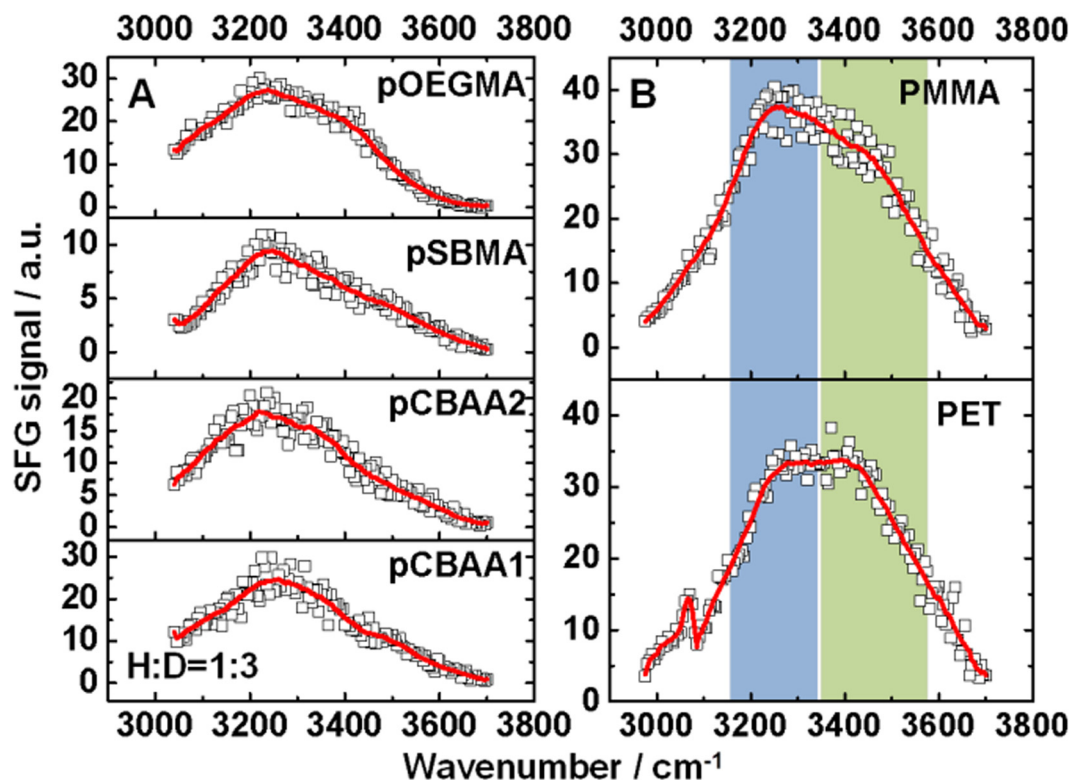


Fig. 5. SFG spectra collected from the isotopically diluted water (H:D = 1:3) in contact with the surfaces of (A) nonfouling pCBAA1, pCBAA2, pSBMA and pOEGMA, and (B) control polymers PET and PMMA [51]. (Adapted with permission from *J. Phys. Chem. C* 119 (2015) 8775–8780. Copyright 2015 American Chemical Society.)

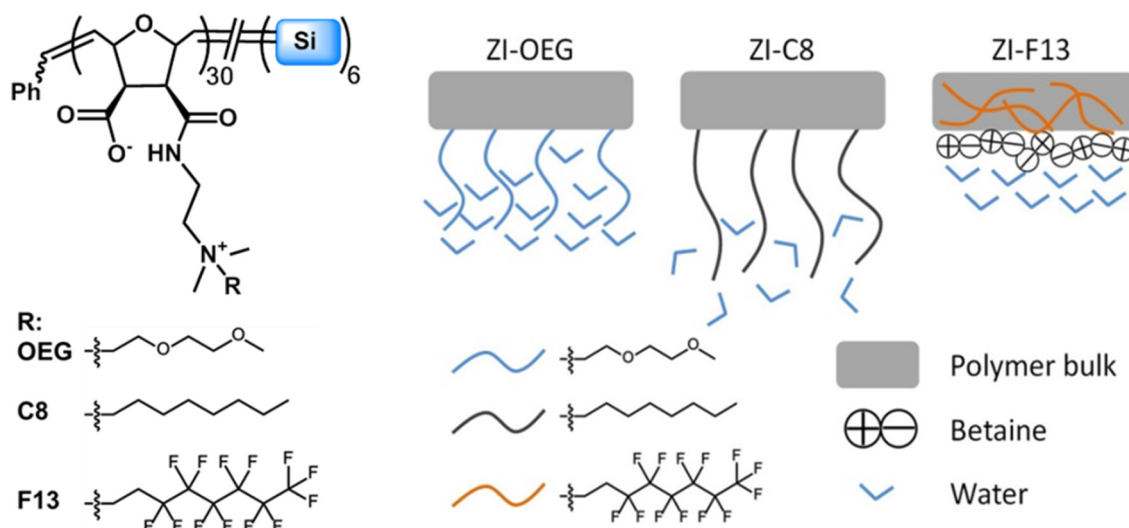


Fig. 6. Molecular structures of amphiphilic polybetaines with different side chains prepared on silica substrates and schematics showing their surface structures in water [53]. (Adapted with permission from *ACS Macro Lett.* 2 (2013) 1011–1015. Copyright 2013 American Chemical Society.)

ens the electrostatically induced hydrogen bonding and reduces the ordering of the interfacial water. As a result of stronger interfacial hydration, for pCBAs, a shorter carbon spacer between the positive and negative charges and a higher environmental pH lead to better nonfouling property as reported in the literature [68,69,71].

3.1.3. Negatively charged functional groups

Different from pCBAs, the water signal detected from the pSBMA/water interface stays constant with pH ranging from 5 to 9, indicating that the interfacial water ordering is unaffected by the change of pH of the aqueous phase in contact with the polymer (Fig. 4C). This is because sulfonic acid is generally a much stronger acid than carboxylic acid. As a result, the sulfonate group in sulfobetaine is harder to be protonated than the carboxylate group in carboxybetaine. In the pH range from 5 to 9, the sulfonate group cannot be protonated, and the ordering of interfacial water stays

the same [52]. This is consistent with a previous study which shows that the nonfouling property of pSBMA is independent of pH [69].

The comparison between pCBAs and pSBMA demonstrates that the strong water signals at 3200 cm^{-1} in SFG spectra are correlated to good nonfouling performance of the zwitterionic materials, indicating that strong ordering of the strongly hydrogen-bonded water molecules (strong hydration) at the materials' surfaces plays a key role in preventing biofouling and achieving nonfouling.

3.1.4. Isotopic dilution

Isotopic dilution, or using HOD instead of H_2O , greatly simplifies the vibrational spectrum of water by reducing the O–H stretching modes and eliminating the Fermi resonance, which resolves the controversy on the assignments of water signals and allows better interpretation of interfacial SFG water spectra [34–36]. Fig. 5

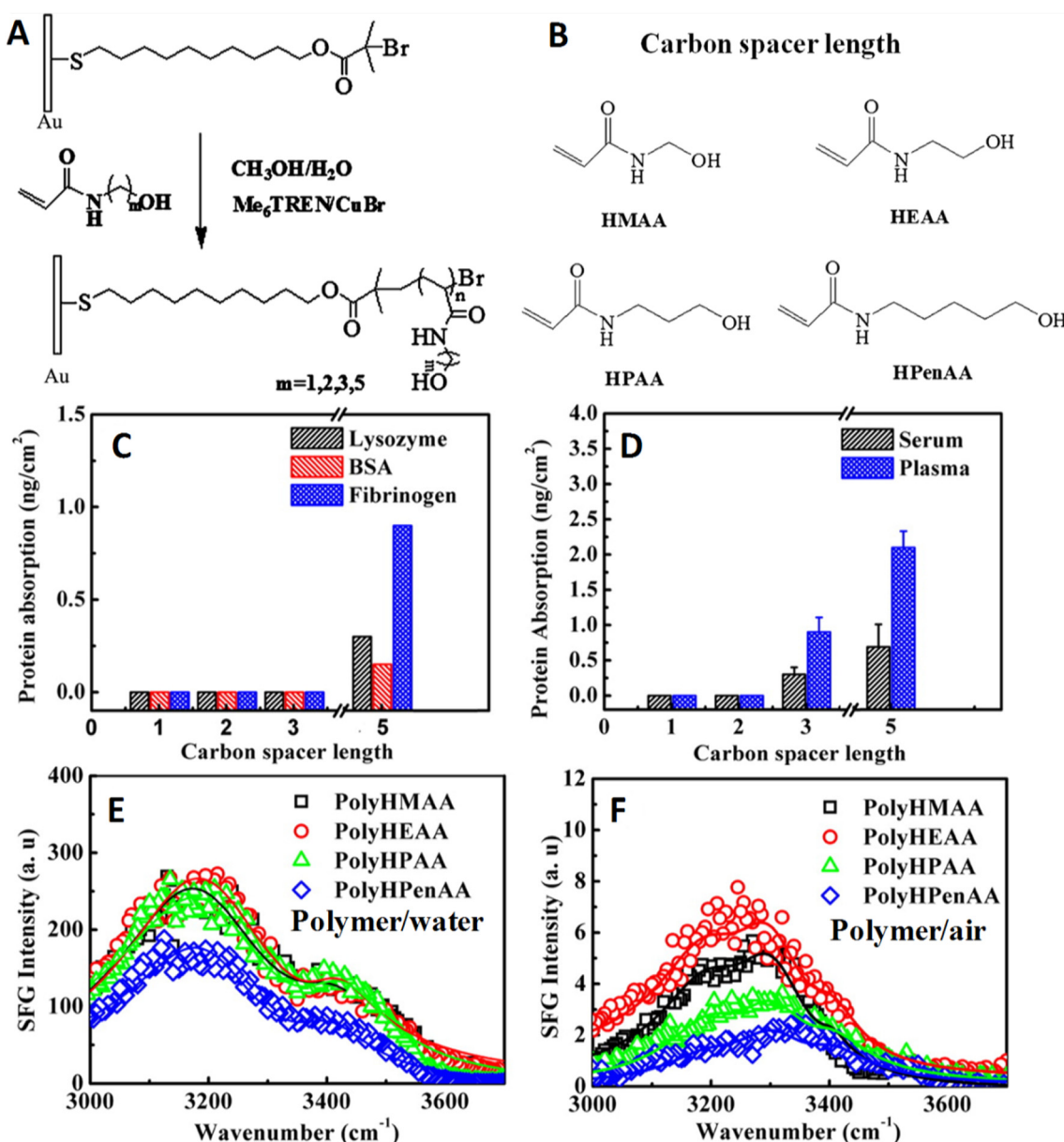


Fig. 7. (A) Preparation of PolyHAAA-based brushes on a gold surface. (B) Molecular structures of HMAA, HEAA, HPAA, and HPenAA monomers with different carbon spacer lengths. Protein absorption amounts from (C) three single protein solutions and (D) undiluted human blood plasma and serum on surfaces of PolyHAAAs with various carbon spacer lengths determined by surface plasmon resonance. SFG spectra of PolyHAAAs (E) in water and (F) in air [74]. (Adapted with permission from *Biomacromolecules* 15 (2014) 2982–2991. Copyright 2014 American Chemical Society.)

shows the SFG water spectra taken at the surfaces of nonfouling zwitterionic polymers and PEG as well as control polymers poly(ethylene terephthalate) (PET) and poly(methyl methacrylate) (PMMA) in contact with isotopically diluted water ($\text{H}_2\text{O}:\text{D}_2\text{O} = 1:3$) [51]. The water spectra at the surfaces of zwitterionic polymers are similar to that of PEG. They are dominated by the strong band at 3200 cm^{-1} contributed from strongly hydrogen-bonded water molecules at interfaces (Fig. 5A). Therefore, strong hydration can be correlated to the nonfouling property of these materials. Different from the nonfouling polymers, for the control polymers, a strong SFG band at 3400 cm^{-1} was observed in the water spectra, indicating a significant amount of weakly hydrogen-bonded water (Fig. 5B). Correspondingly, these polymers are not antifouling. In the isotopic dilution study, we again demonstrated that the strongly hydrogen bonded water or strong hydration is important for nonfouling.

3.2. Amphiphilic polybetaines

The above zwitterionic polymers are based on acrylate or methacrylate backbones. Zwitterionic functional groups can be integrated into polymers with different backbone structures, as shown in Fig. 6, denoted as ZI-OEG, ZI-C8, and ZI-F13 [53]. It has been shown that such polymers containing side chains with different hydrophilicities exhibit varied antifouling properties [18,72]. We believe that this is because the different side chains lead to different surface structural changes in water. The surface/interfacial structures of the polymers (e.g., presence of certain functional groups, their coverage, orientation, and ordering) were deduced from the SFG signals from specific functional groups of the polymers. First we performed SFG studies on these polymers in air. The side chains of the three polymers are all present at the surfaces in air [53]. We then investigated the molecular surface structures of these polymers in water *in situ*. In water, the oligo(ethylene glycol) (OEG) and alkyl (C8) side chains remain on the surface, while the fluorocarbon (F13) chains bend into the polymer bulk away from water because of their unfavorable interactions with water. For ZI-F13, both the zwitterionic groups are present at the water interface, where the carboxylate groups form hydrogen bonds with water molecules.

The antifouling mechanisms of the above amphiphilic polybetaines can be well explained by their surface restructuring behaviors. The good antifouling property of ZI-OEG is due to the presence of the OEG groups on the surface in water. It has been extensively shown that the surface OEG groups lead to improved antifouling activity due to strong hydration [10–12]. The poor antifouling performance of the ZI-C8 is due to the presence of the C8 side chains at the water interface, which have weak hydration and lead to protein adsorption via hydrophobic interactions. The ZI-F13 shows good antifouling performance, because the F13 side chains retreat from water into the polymer bulk, allowing the exposure of the zwitterionic groups to the water interface, which form strong hydrogen-bonds with water molecules. This research indicates that side chains can greatly influence the polymer surface structures in water and their hydration, resulting in different antifouling properties. This research also demonstrates that interfacial hydration is important to ensure the excellent antifouling performance.

4. PolyHAAA-based brushes

We believe that the antifouling performance of other hydrophilic or amphiphilic materials besides zwitterionic polymers should also be influenced by their surface hydration [73]. A series of hydrophilic PolyHAAA-based brushes were prepared on gold substrates via ATRP (Fig. 7A) with varied carbon spacer lengths of the side chains (Fig. 7B). All the PolyHAAAs studied here show excellent antifouling properties, especially, PolyHAAAs with shorter carbon spacer lengths exhibit even better antifouling abilities in both protein solutions (Fig. 7C) and undiluted human blood serum/plasma (Fig. 7D). SFG results further revealed that PolyHMAA and PolyHEAA, both with short carbon spacer lengths of the side chains, can retain and hold interfacial water more effectively than PolyHPMA and PolyHPenAA in both wet (Fig. 7E) and dry (Fig. 7F) states, consistent with the enhanced antifouling ability of PolyHMAA and PolyHEAA. These data again suggest that the antifouling performance of the hydrophilic polymers is positively correlated with their hydration property. The increase of carbon spacer length between the amide and the hydroxyl groups reduces binding affinity between the interfacial water molecules and the

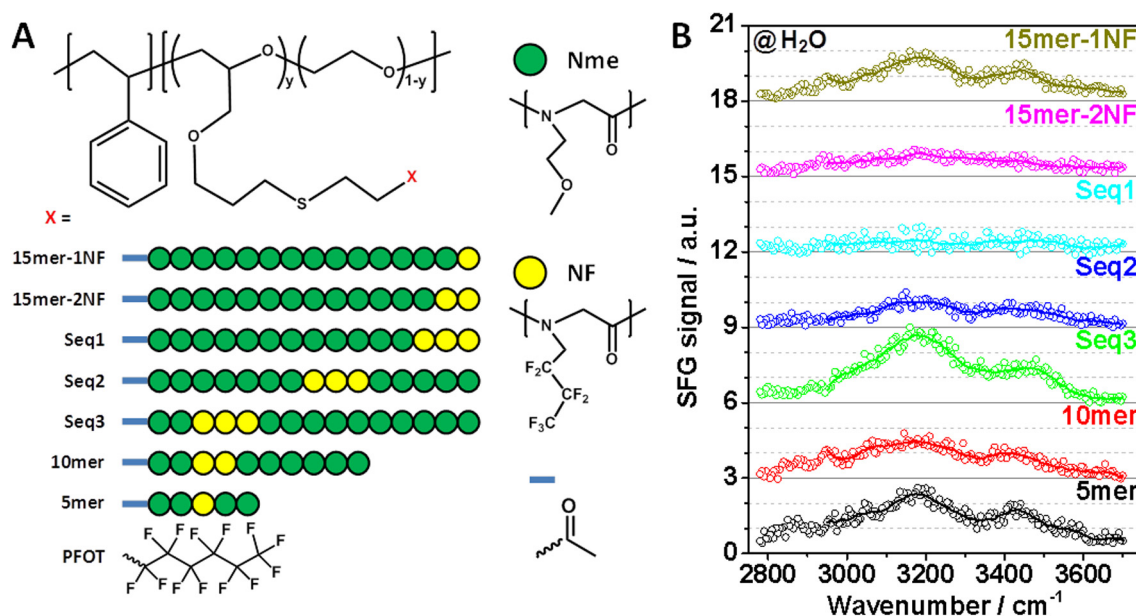


Fig. 8. (A) Molecular structures of sequence-specific amphiphilic polypeptoids. (B) SFG spectra collected from various polypeptoid coatings in water [57]. (Adapted with permission from *Langmuir* 31 (2015) 9306–9311. Copyright 2015 American Chemical Society.)

polymer brushes studied here, which weakens their antifouling performance [74].

5. Polypeptoids

Polypeptoids (or N-substituted glycines) have emerged as a class of interesting materials due to their sequence-specificity [75–78]. Hydrophilic and hydrophobic peptoid units can be incorporated into one polymer by rational design of an amphiphilic polymer, the antifouling property of which can be tuned by a subtle change in the number or position of the structural units [79]. The molecular structures and name abbreviations of several such amphiphilic polypeptoids with systematically varied sequences are shown in Fig. 8A. We performed SFG studies on these polymers to understand the correlations among their surface hydration, molecular formula, and antifouling behaviors [57]. The SFG spectra of the polymer/water interfaces (Fig. 8B) indicate that the ability of the polymer surfaces to orient and form strongly hydrogen-bonded water is indeed sensitively dependent upon the sequence of the peptoids. Combined with their antifouling test results [79], we conclude that strong interactions between the polymers and water molecules are well-correlated to their antifouling properties. Specifically, the best antifouling performance was observed on the Seq3 coating [79], which also showed the strongest hydration by SFG. Notice that a shoulder peak at 3400 cm^{-1} assigned to weakly hydrogen bonded water was also observed. For antifouling applications, these materials may not have been completely optimized, but the trend concerning hydration and antifouling property can be well-established.

6. Interactions between proteins and antifouling polymers

As nonfouling polymers, pSBMA and pOEGMA perform similarly well in resisting protein adsorption, and they both are strongly hydrated while in contact with water [51]. However, we found that their hydration is affected at different degrees by protein contact [50]. The SFG spectra of the pSBMA/protein solution interfaces are identical to those taken at the pSBMA/water interface after water was replaced by the protein solutions, indicating that the interfacial water molecules strongly bonded with pSBMA cannot be disturbed by the protein molecules (Fig. 9A). Different from pSBMA, the water signal at the pOEGMA surface became either stronger or weaker upon contacting protein solutions depending on the type of the proteins, indicating a noticeable impact of the presence of proteins on the water structure at the pOEGMA surface (Fig. 9B). Nevertheless, the water signal at the pOEGMA surface was completely recovered when the surface was rinsed and placed in pure water again, consistent with its nonfouling property. In contrast to nonfouling pSBMA and pOEGMA, on a silica surface, the water signal cannot recover after contacting proteins, indicating irreversible protein adsorption.

The impact of sulfobetaine and PEG on the surface hydration of proteins was also investigated by SFG (Fig. 9C–E). Whereas water was strongly hydrogen bonded at the protein surfaces with the presence of free sulfobetaine (SB) molecules, weakly hydrogen bonded or disordered water molecules were found at the interfaces between proteins and free PEG solutions. Different from free PEG molecules, when the proteins were in contact with a PEG-coated gold nanoparticle solution, strongly hydrogen-bonded water was

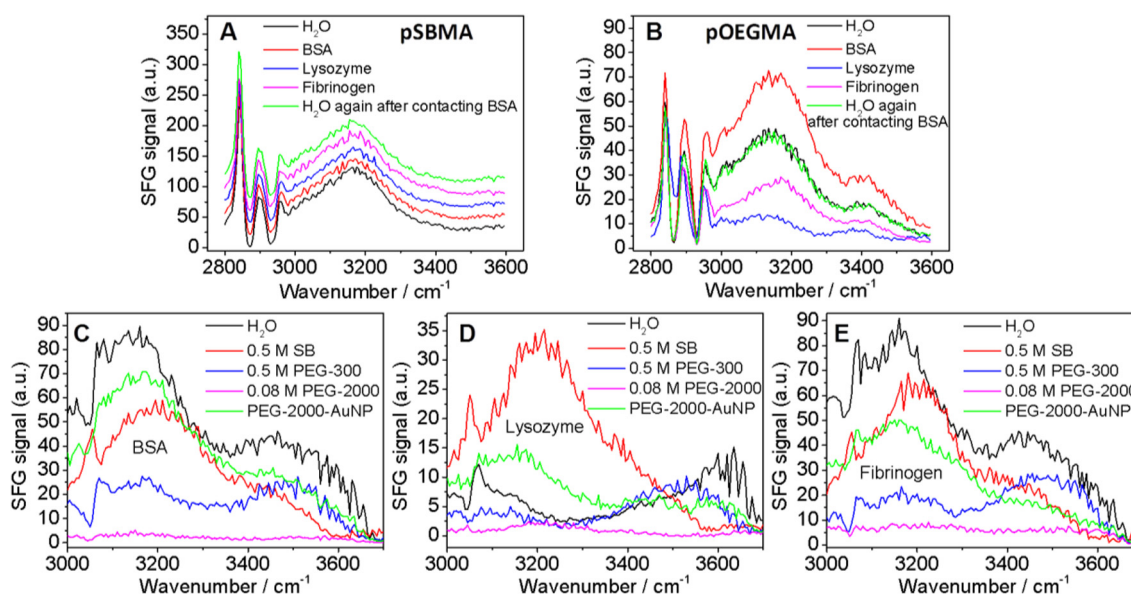


Fig. 9. SFG spectra of (A) pSBMA and (B) pOEGMA in contact with water and various protein solutions (1 mg/mL), and of (C) bovine serum albumin (BSA), (D) lysozyme, and (E) fibrinogen surfaces in contact with water and aqueous solutions of sulfobetaine (SB) (0.5 M), PEG-300 (0.5 M), PEG-2000 (0.08 M), and PEG-2000 coated gold nanoparticles (OD = 5) [50]. (Adapted with permission from *ACS Appl. Mater. Interf.* 7 (2015) 16881–16888. Copyright 2015 American Chemical Society.)

Table 1

Antifouling ability, hydration, and protein interaction of antifouling materials and model control surfaces.

	Samples	Antifouling	Hydration	Proteins
Antifouling material candidates	Zwitterionic	Yes	Strong	Strong hydration
	PEG	Yes	Strong	Strong hydration
	PolyHAAA	Yes	Strong	
	Amphiphilic	Yes	Strong	
Control surfaces	PMMA, PET	No	Weakly H-bonded water	
	SiO ₂	No		Weakly H-bonded water

observed to dominate at the interface, which is similar to the interface between pOEGMA and protein solutions. The PEG chains tethered on a surface are closely packed and have limited freedom to change conformation to directly interact with proteins. Also strong hydration shells are formed on top of tethered PEG, which further prevent PEG to interact with proteins.

The above SFG study on the surface hydration of both polymers and proteins suggests that for PEG-coated surfaces including nanoparticles, although surface hydration may be disrupted by proteins to some extent, it is still relatively strong and resists protein adsorption. However, free PEG binds to proteins, reducing their hydration significantly. For SB, independent of its state in water (free or bound), its surface hydration remains strong in contact with proteins [50]. This study further demonstrated that for zwitterionic polymers and surface immobilized PEGs, the excellent nonfouling property is resulted from strong hydration.

7. Conclusions and perspectives

Here we showed that SFG spectroscopy has been successfully applied to study the surface hydration of antifouling polymers. The antifouling polymers examined include zwitterionic polymer brushes with various molecular structures, uncharged hydrophilic polymers, and amphiphilic polymers. Their antifouling ability, surface hydration, and interactions with proteins have been examined and compared with control surfaces, as summarized in Table 1. One general discovery in these SFG studies is the correlation between strongly hydrogen bonded water molecules at the polymer surfaces and the excellent antifouling performance of these polymers. This reveals a general design rule for effective antifouling materials.

SFG spectroscopy is a powerful technique to study solid/liquid interfaces to elucidate surface hydration. Novel antifouling materials are emerging with unprecedented compositions and properties, such as self-healing zwitterionic hydrogels [80], omniphobic slippery surfaces [81], xerogels [82], etc, SFG spectroscopy will be applied to reveal the surface structure, hydration, and protein interactions of these new polymers in the future. In addition, SFG has been proved to be powerful to reveal surface/interfacial structures under flowing water [83], and because the antifouling materials are expected to be applied in contact with flowing water, it will be interesting to reveal the effect of water flow on the surface structures and hydration of antifouling materials. We believe that SFG spectroscopy will continue to be influential to provide molecular insights into new antifouling mechanisms and guide the design of new materials with improved properties.

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